



MARCH 07 - MARCH 10, 2026
WASHINGTON HILTON | WASHINGTON, DC

Better late than never:

*Late presenting double jeopardy AMI-CS
successfully treated with multivessel PCI*

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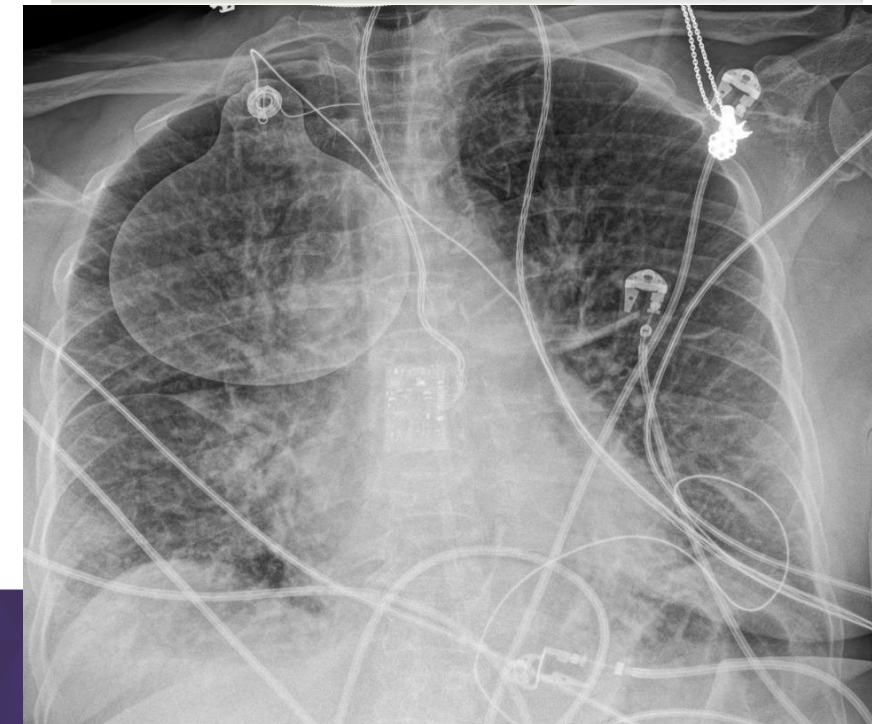
Jacob McAuliffe, MD

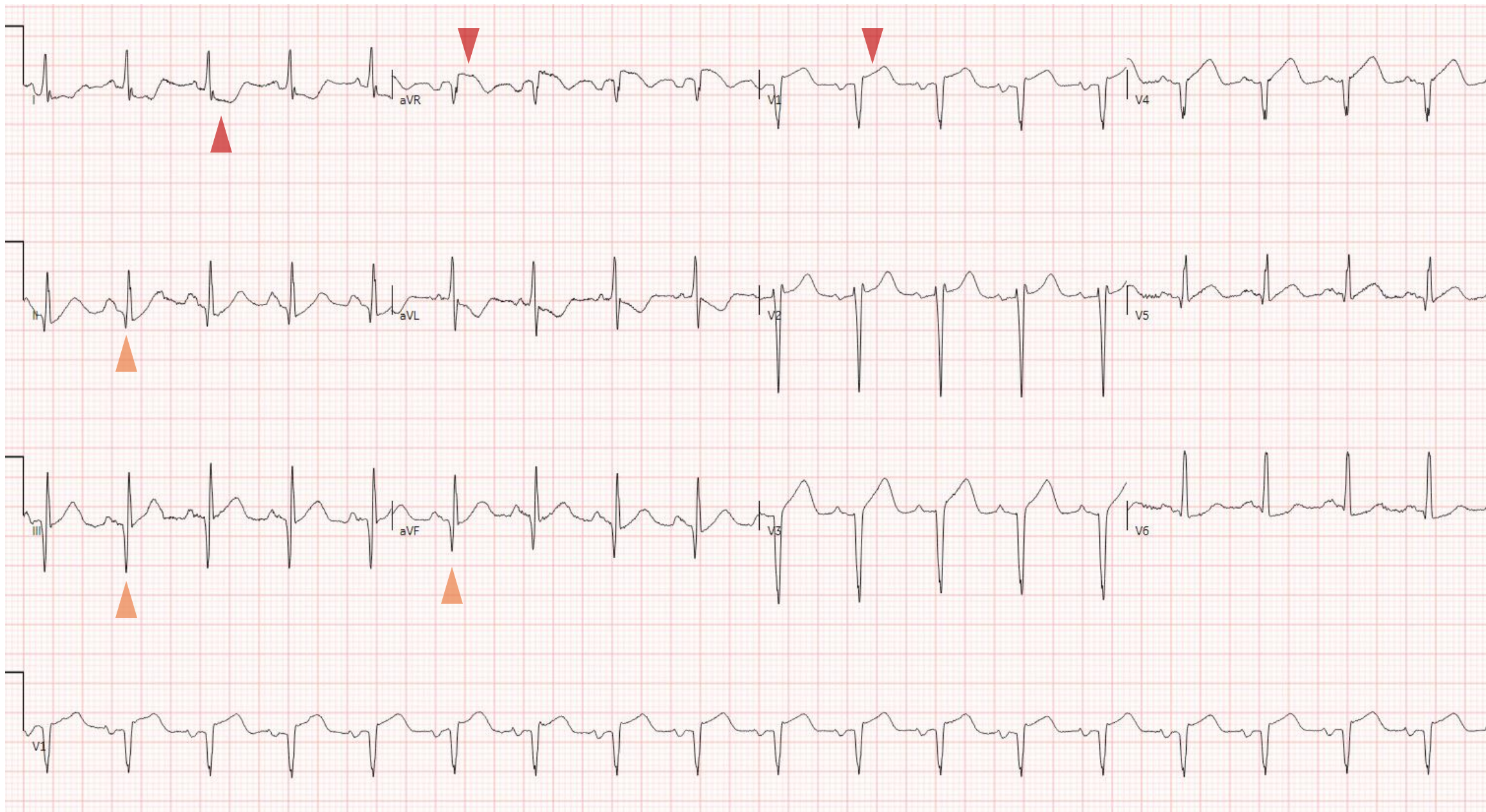
- I have no relevant financial relationships

Case Presentation:

- 77 yo female with only known PMH of HTN
- Presented ~3:45 am through OSH ED with 14 hrs of acute onset substernal chest pain
 - Radiates to epigastrium and upper back; associated with rapidly worsening dyspnea, diaphoresis, and nausea
- Never smoker; FHx father with MI in 70s
- Previously active able to tolerate swimming multiple days per week

	Range & Units	03:53	04:14
WBC	3.10 - 9.50 x10 ³ /uL	16.26 (H)	
Hemoglobin	11.4 - 14.8 g/dL	11.5	
Platelet	142 - 346 x10 ³ /uL	206	
Creatinine	0.4 - 1.0 mg/dL	0.7	
Sodium	135 - 145 mEq/L	134 (L)	
Potassium	3.5 - 5.3 mEq/L	5.2	
EGFR	>=60.0 mL/min/1.73 m ²	>60.0	
Lactate	0.4 - 2.0 mmol/L		
AST	<=41 U/L	118 (H)	
ALT	<=55 U/L	78 (H)	
Bilirubin Total	0.2 - 1.2 mg/dL	0.5	
hs Troponin-I	<=14.0 ng/L		4,484.5
NT-proBNP	<=956.1 pg/mL		6,258.8





Emergent Coronary Angiography

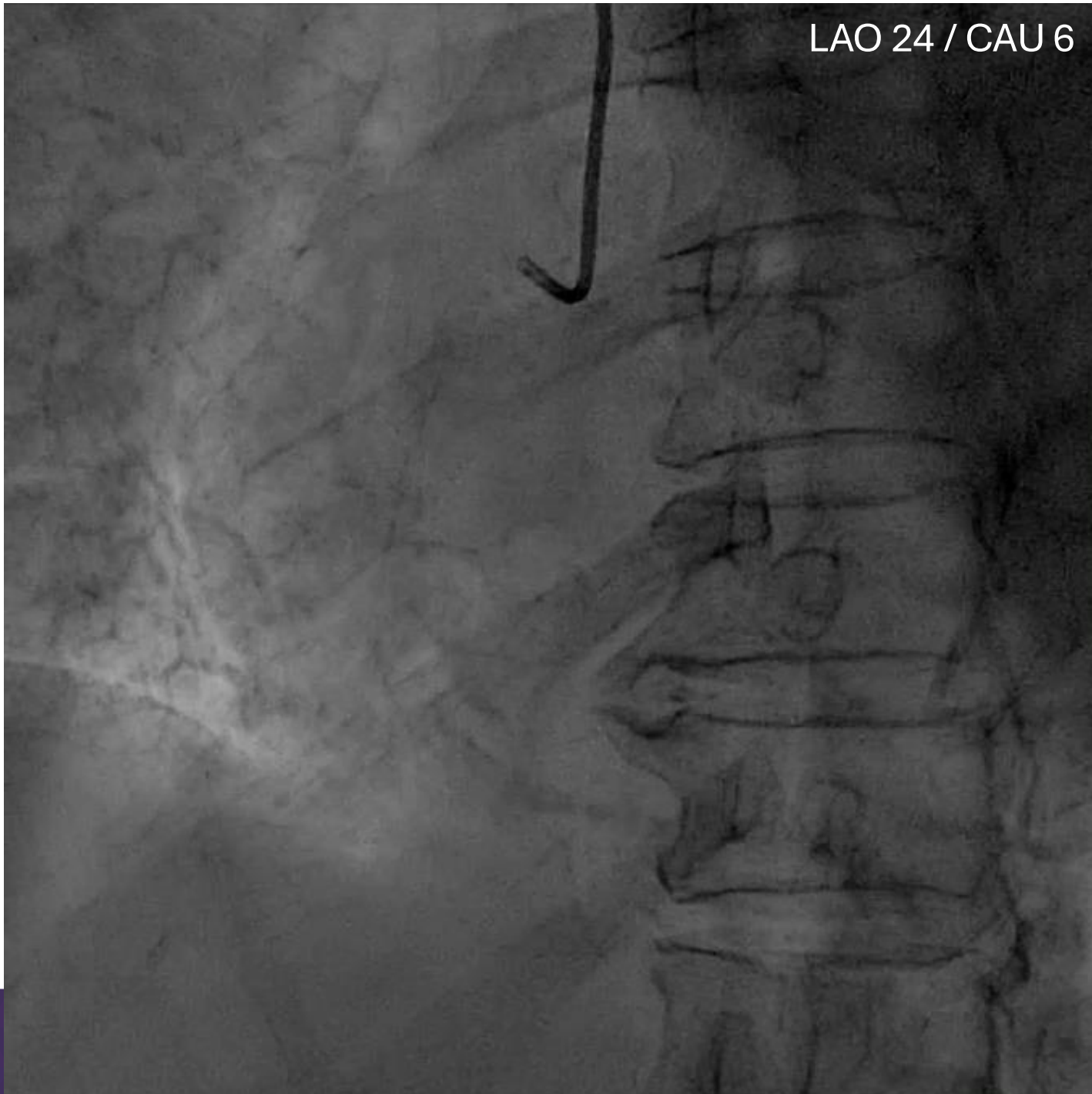
- prox LCx subtotal occlusion; moderate diffuse mid disease; large tandem OMs
- ostial LAD subtotal occlusion; severe bifurcation lesion at large first septal perforator
- Prominent L -> R collaterals
- significant calcification

Emergent Coronary Angiography

- prox LCx subtotal occlusion; moderate diffuse disease distally
- ostial LAD subtotal occlusion; severe bifurcation lesion at large first septal perforator
- Prominent L -> R collaterals
- significant calcification

Emergent Coronary Angiography

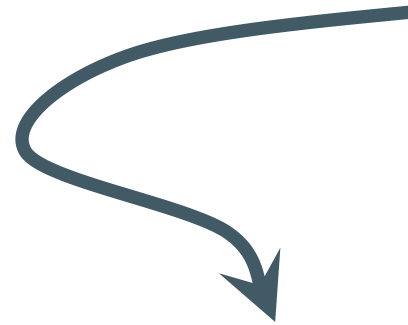
- RCA CTO
- R -> R & L collaterals



RHC

- Elevated b/l filling pressures
- Low cardiac output
- Elevated LVEDP, low CPO
- *Severe triple vessel disease*

RA	20
RV	59/15
PA	66/44, mean 56
PCW	41
Fick	2.6 (1.7)
TD	2.4 (1.6)
PVR	6
LVEDP	36
Sys BP	96/63 (74)
CPO	0.41
PAPi	1.1



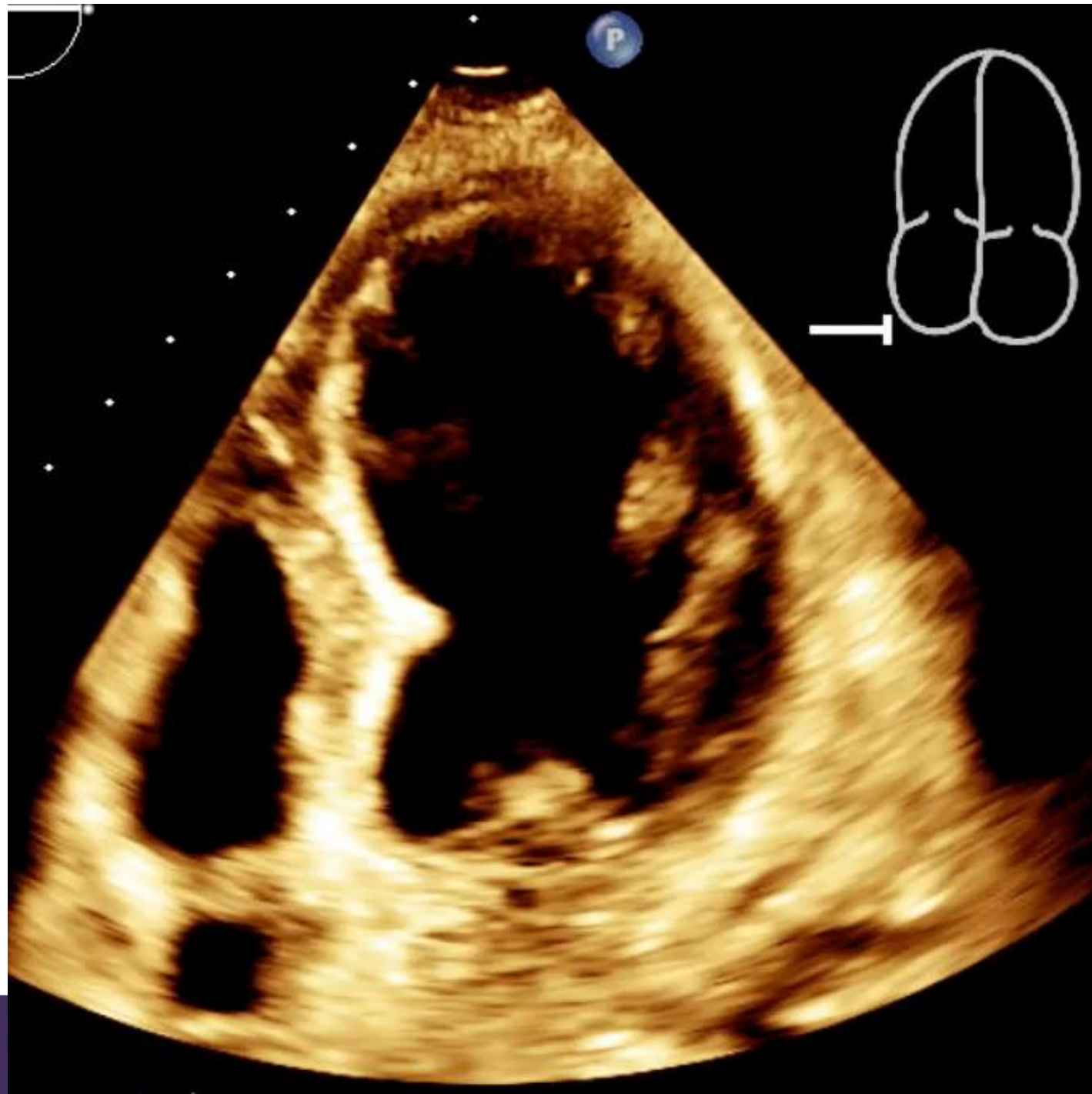
	Range & Units	03:53	04:14	08:34	09:14	12:21
WBC	3.10 - 9.50 x10 ³ /uL	16.26 (H)				
Hemoglobin	11.4 - 14.8 g/dL	11.5		12.0		11.9
Platelet	142 - 346 x10 ³ /uL	206				
Creatinine	0.4 - 1.0 mg/dL	0.7		0.7		
Sodium	135 - 145 mEq/L	134 (L)		136		
Potassium	3.5 - 5.3 mEq/L	5.2		4.2		
EGFR	>=60.0 mL/min/1.73 m ²	>60.0		>60.0		
Lactate	0.4 - 2.0 mmol/L				1.5	
AST	<=41 U/L	118 (H)		141 (H)		
ALT	<=55 U/L	78 (H)		79 (H)		
Bilirubin Total	0.2 - 1.2 mg/dL	0.5		0.6		
hs Troponin-I	<=14.0 ng/L		4,484.5	15,131.8		57,036.8
NT-proBNP	<=956.1 pg/mL		6,258.8			

Admitted to CICU for BiV AMI-CS, SCAI B

- Urgent CABG consult – adamant against CABG
- IABP placed few hours later, improving HFNC req
- Loaded with brilinta
- Gtts: heparin, lasix (10 mg/hr), nipride (intermittent req)

TTE on IABP 1:1

- EF 20-25%
- septal aknetic
- anterolateral dyskinesia with hinge point
- normal RV size/function
- c/f completed LAD infarct



What would you do next?

- Severe 3V disease
 - LAD w/ possibly completed infarct based on TTE & ECG
 - Severe prox LCx disease
 - RCA CTO
- AMI-CS, SCAI C
 - Well supported on IABP 1:1
- ICM w/ HFrEF (20-25%)

1. CABG
(strong patient preference against)
2. High risk PCI
(revascularization targets / strategy?)
3. Continue or escalate tMCS as bridge to decision for VAD/OHT evaluation
(see 1 ^)
4. Medical management
5. Refer to palliative/hospice care

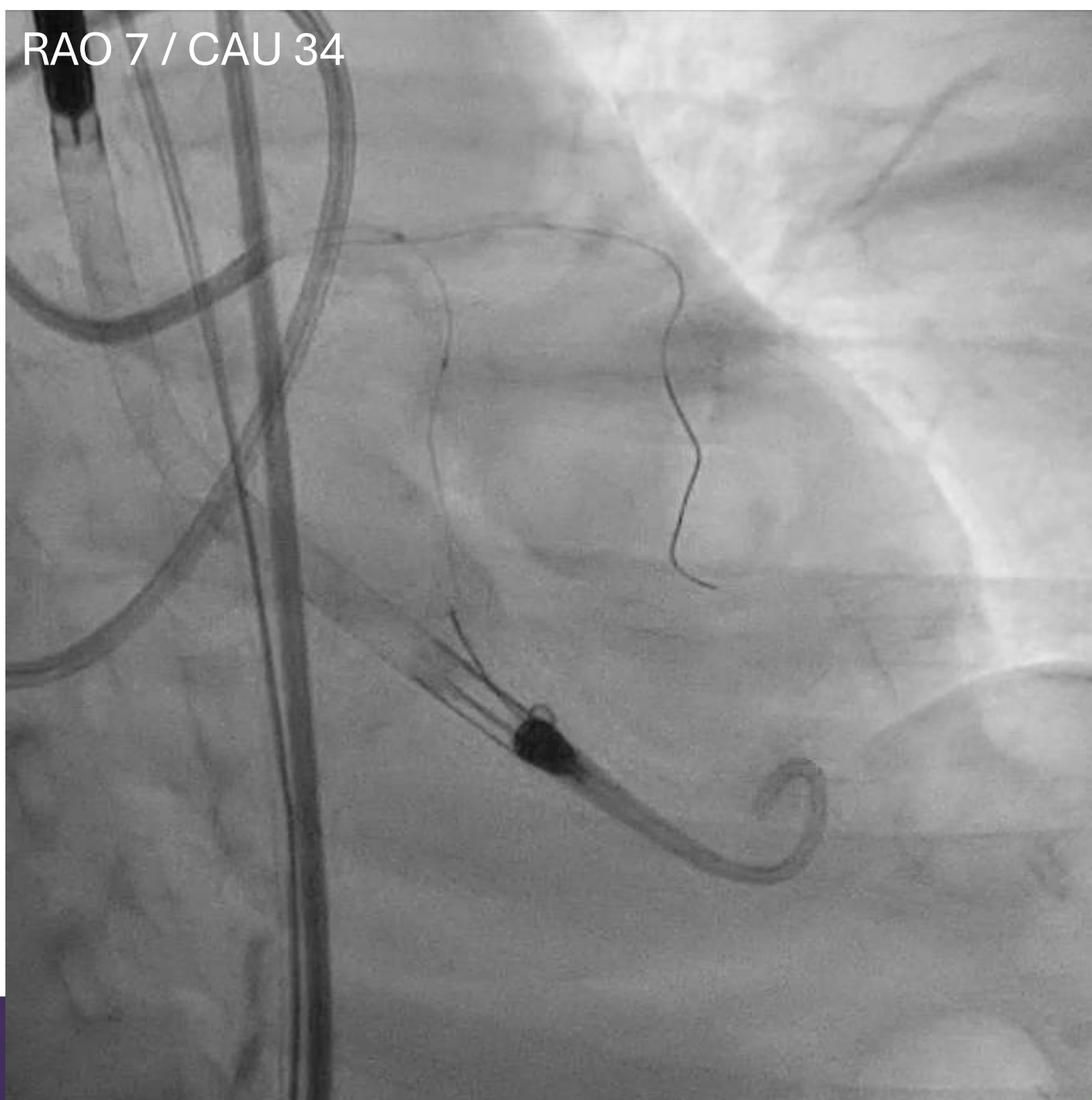
CHIP Planning

- Shared decision making regarding very high-risk nature of procedure
 - quoted as >20% risk of MACE or other major adverse event
- Planned IABP exchange for periprocedural Impella CP support
- Upfront femoral central venous access
- Prioritizing targets:
 1. LM-LCx PCI/DES (“last remaining vessel”, large territory)
 2. RCA CTO
 3. Ostial–?? LAD
- Elective intubation in CICU prior to procedure
- Peripheral vascular screening
- Pre-closure utilized for large bore access
- Cr stable at 0.7 to 1.0

High risk, complex **PCI**

- PA-C and Impella CP in place
- Workhorse wires down LAD and LCx
- No flow to distal LAD
- TIMI 1 LCx

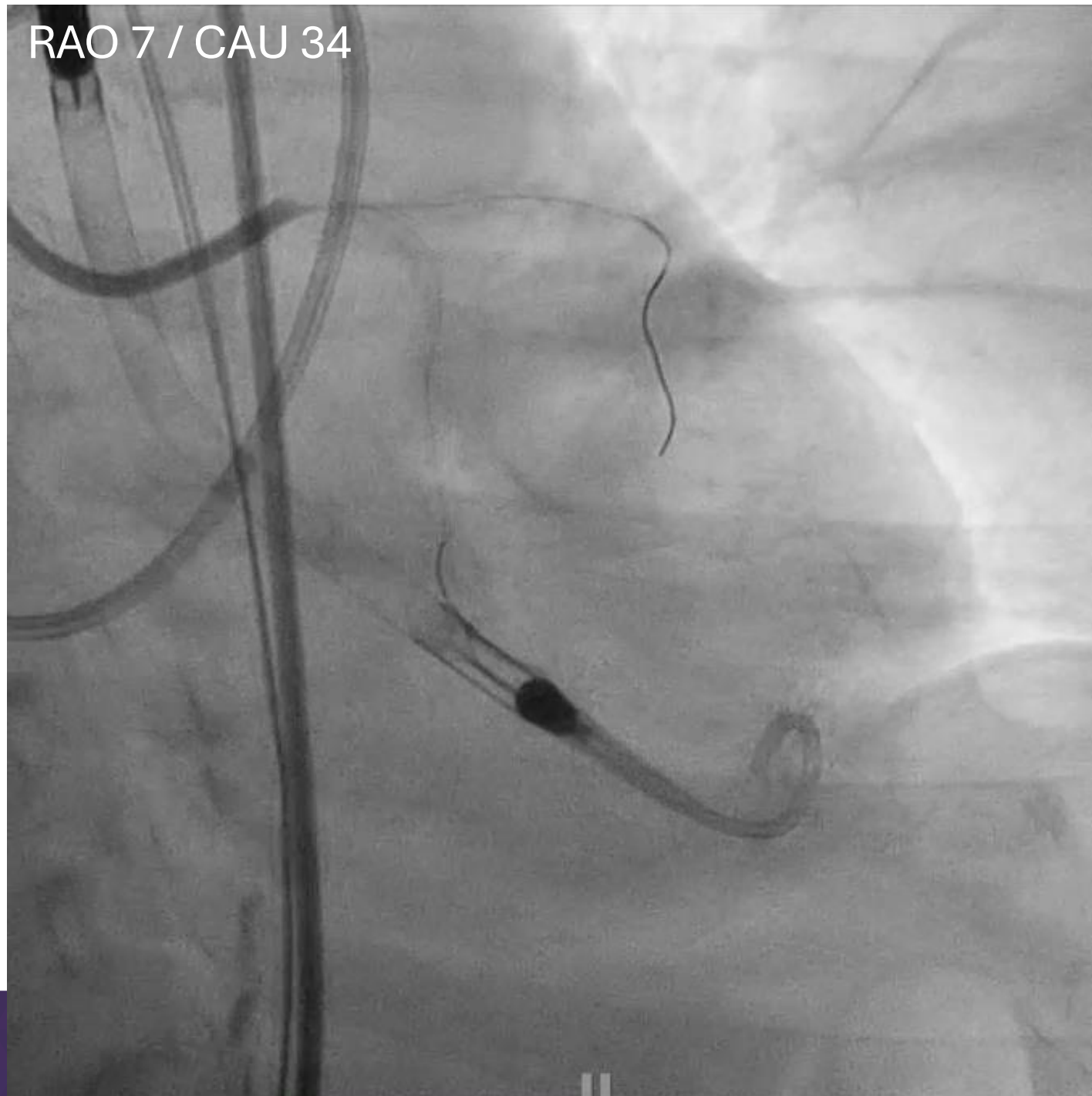
RAO 7 / CAU 34



High risk, complex **PCI**

- Post ostial LCx POBA
- TIMI 3 flow restored

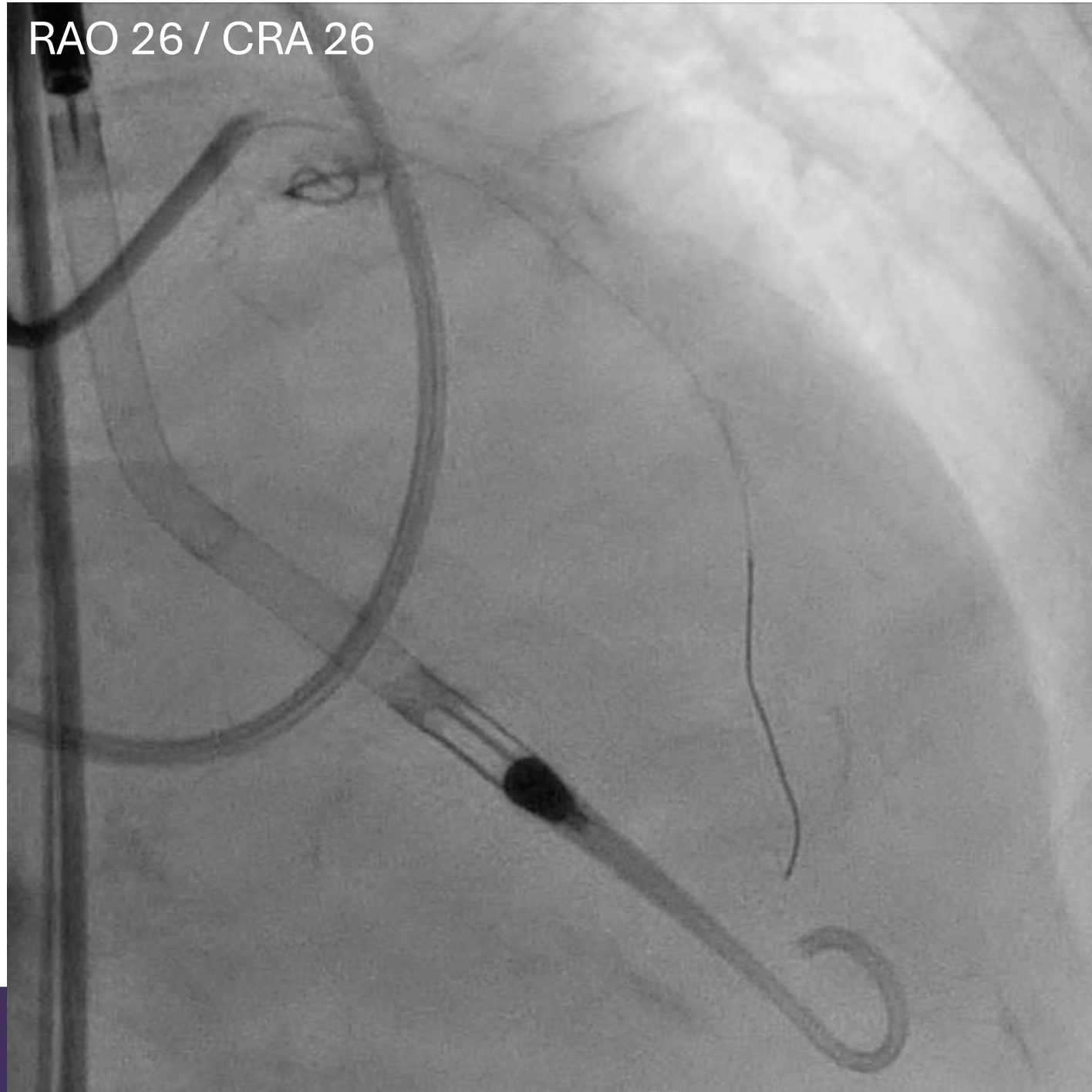
RAO 7 / CAU 34



High risk, complex PCI

- Large septal perforator
- Dependent RCA
- Viable septal base?
- Limited PCI options for LM-LCx DES (primary target) **and** ostial LAD

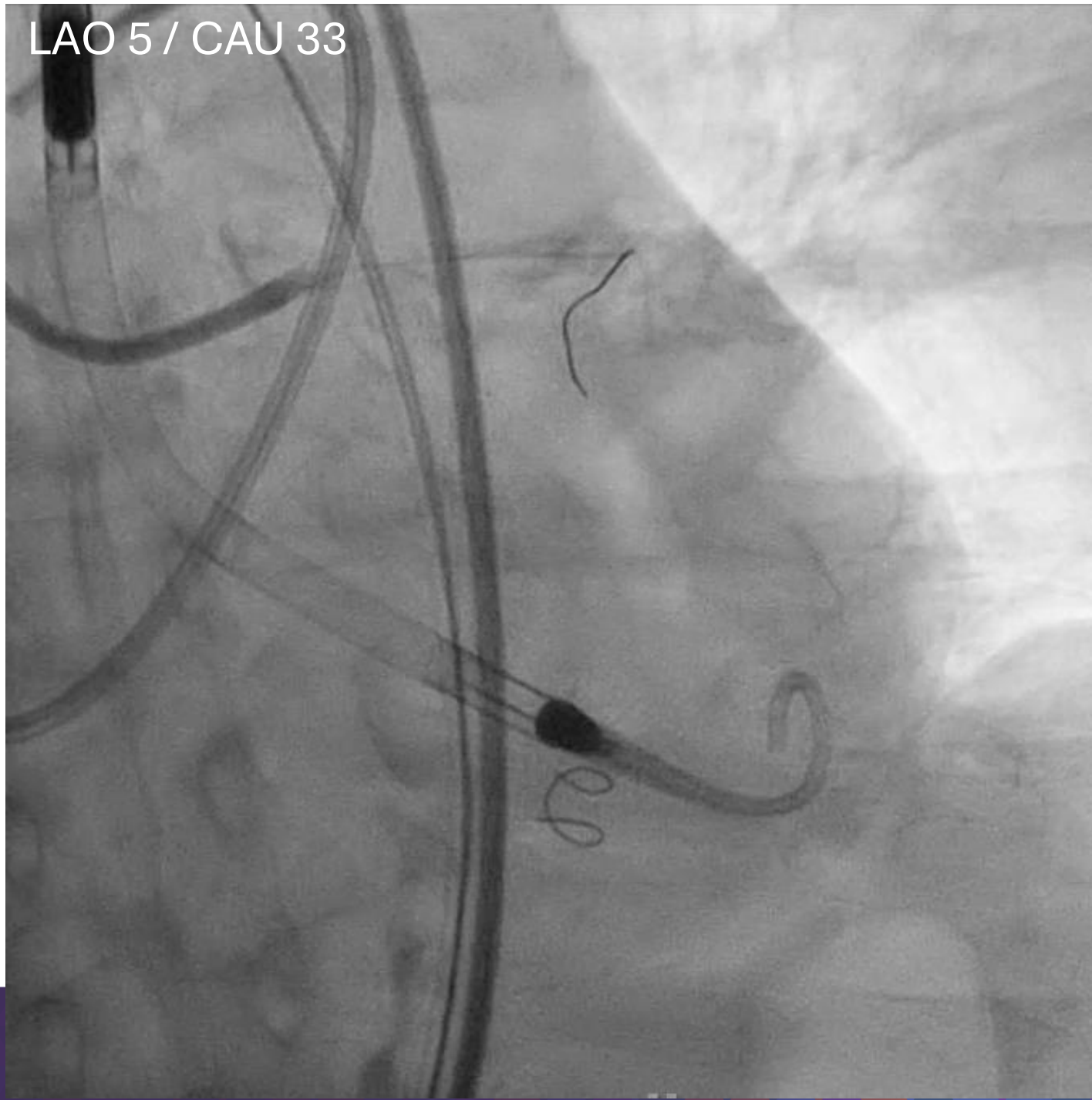
RAO 26 / CRA 26



High risk, complex **PCI**

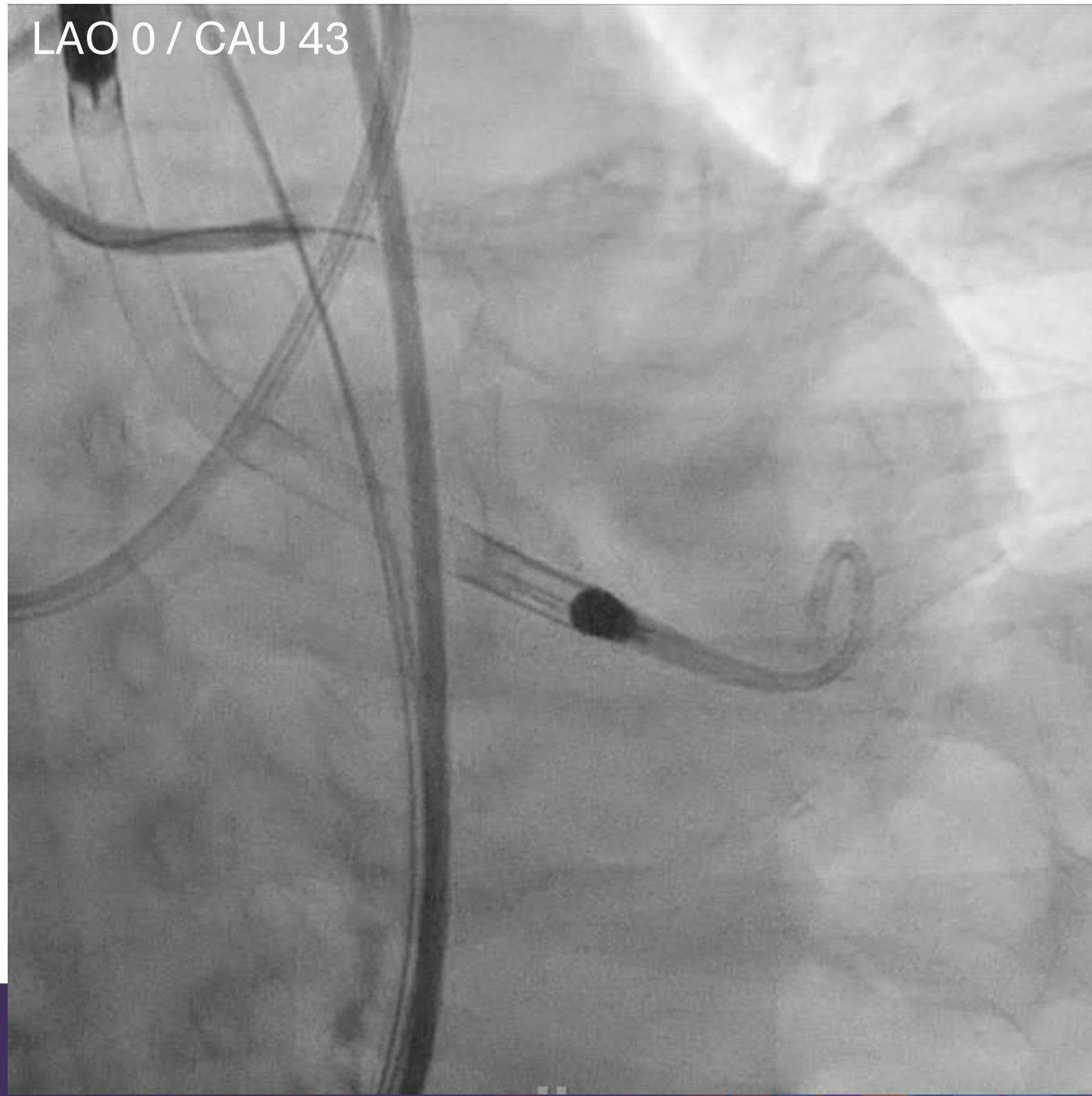
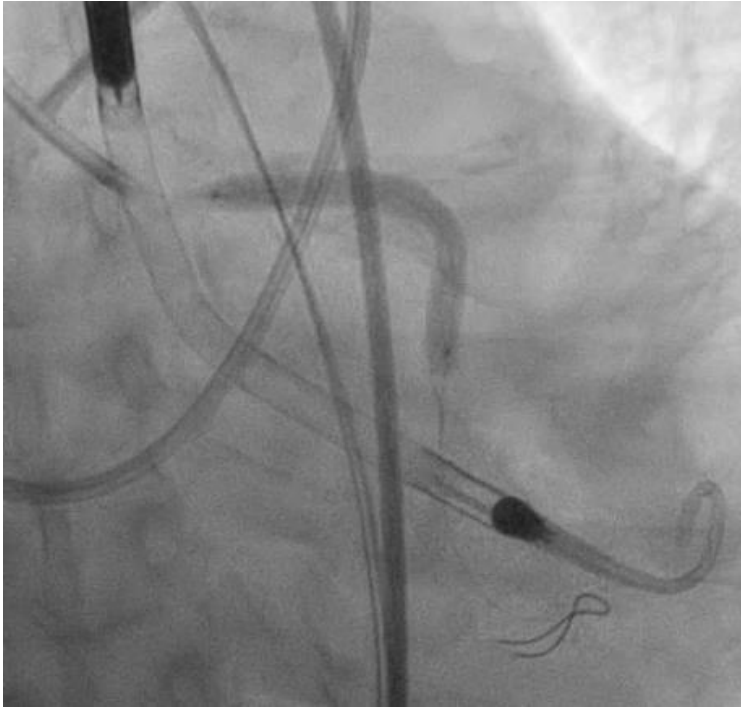
- IVUS-guided POBA + DCB ostial to mid LAD
- Good angiographic result in proximal segment

LAO 5 / CAU 33



High risk, complex **PCI**

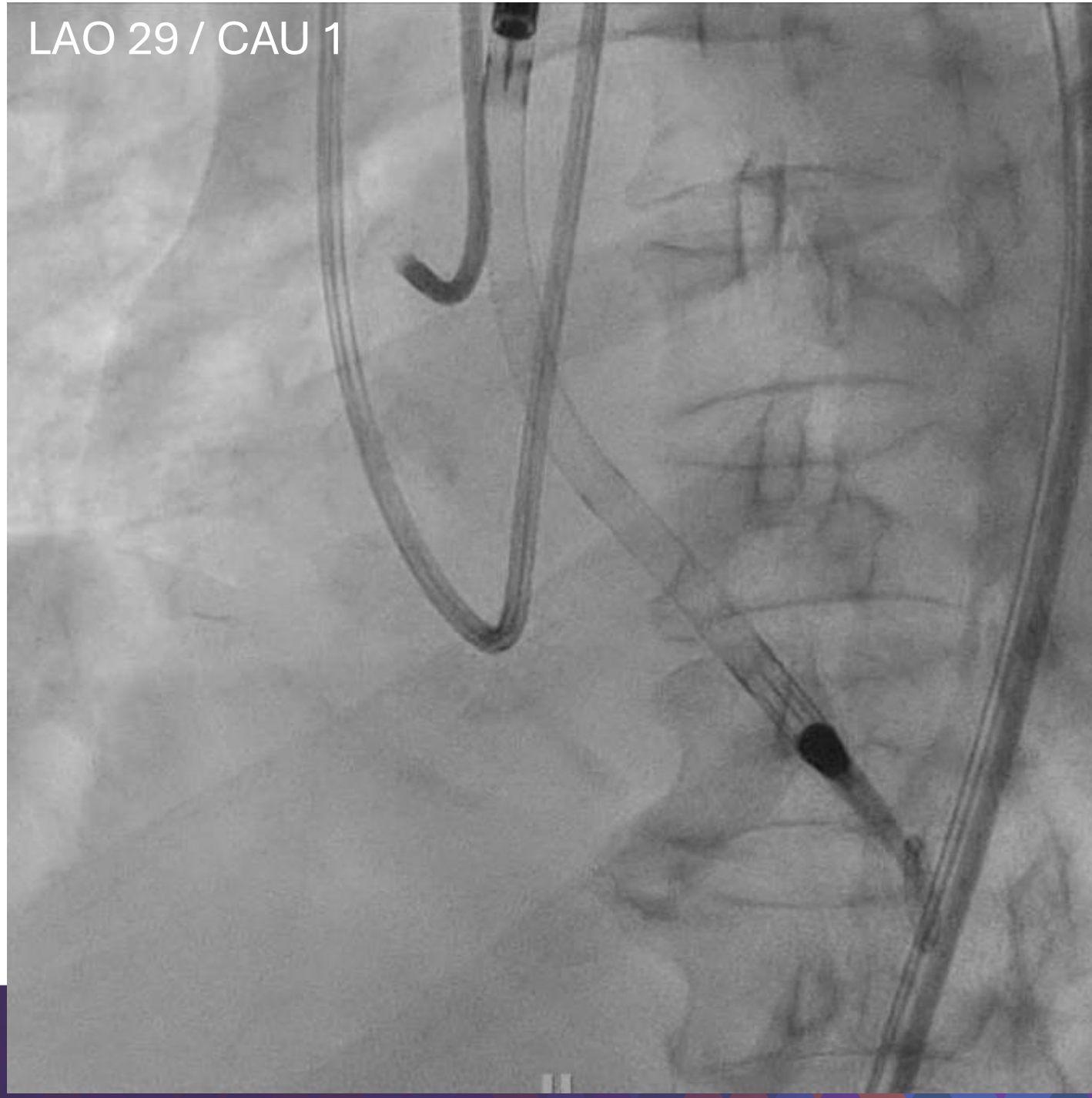
- IVUS-guided LM-LCx DES



High risk, complex **PCI**

- Satisfied with achieved L system revascularization
- Conservative contrast use so far
- Stable hemodynamics

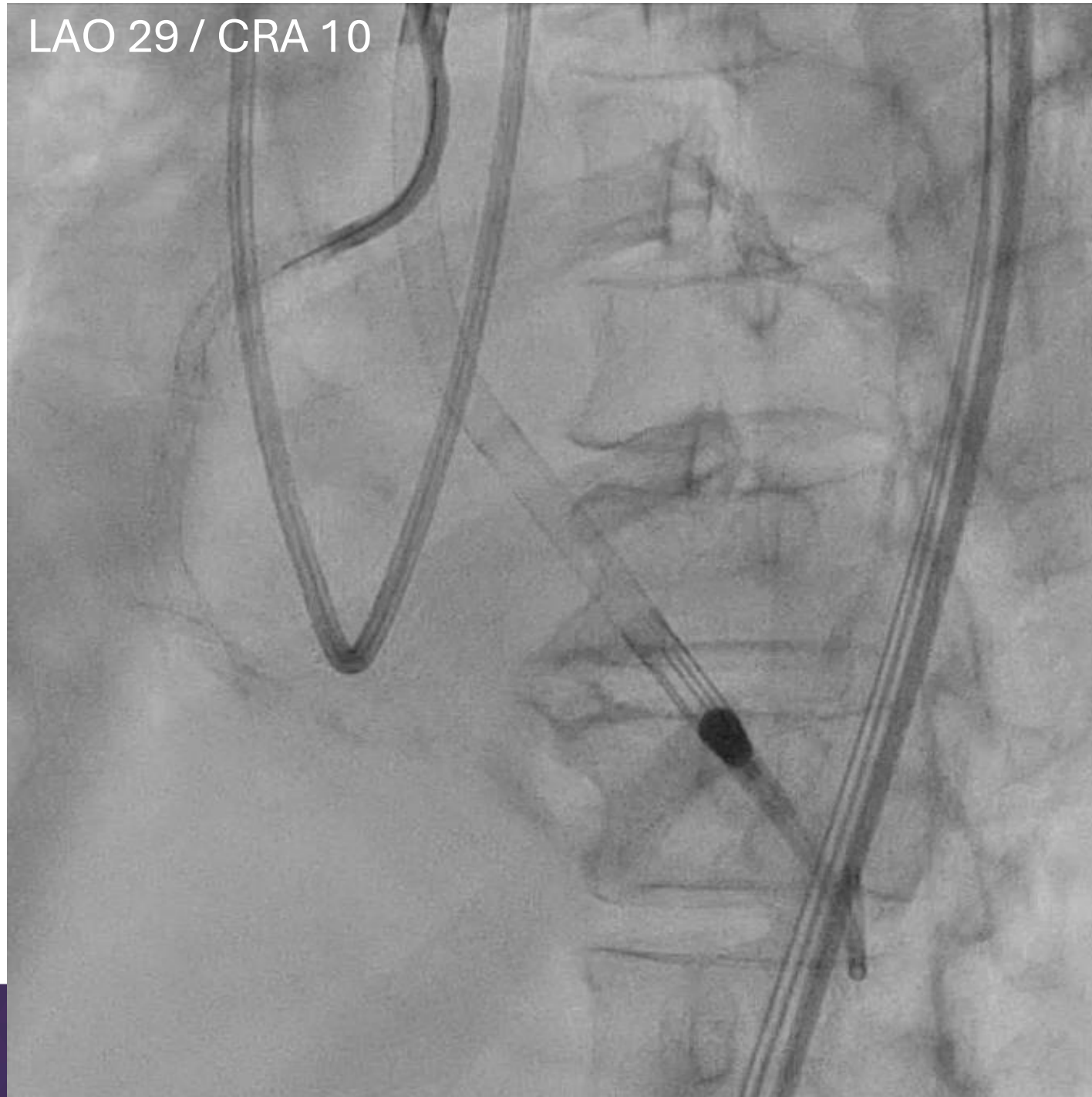
LAO 29 / CAU 1



High risk, complex **PCI**

- Successful CTO traversal on second attempt / device
 - Turnpike; Mongo
- IVUS-guided DES ostial to mid RCA

LAO 29 / CRA 10



Post-procedural course

- Impella CP removed at end of case
- Returned to CICU intubated; extubated following morning
- Downgraded to floor POD 2
- Repeat TTE with EF recovery (20-25% -> 30-35%), similar WMA
- Discharged home POD 4
 - full GDMT regimen w/ DAPT (Brilinta + ASA) and Lipitor 80 mg
- Next steps:
 - cardiac rehab, GDMT optimization, surveillance TTE
 - consider viability study for staged LAD revascularization

Take Home Messages

- **Double jeopardy myocardial infarction**
 - Rare clinical scenario, often requiring complex revascularization efforts
 - Maintain high upfront clinical suspicion when ischemic ECG changes are present across multiple territories
- **Prioritizing revascularization targets, procedural planning**
 - Set your self up for success, adapt to new information and dynamic circumstances
- **Efficacy of regional shock system and Heart Team based care**
 - Appropriate care escalation as patient was transitioning from smoldering SCAI B -> C
 - Early engagement, frequent reassessment

Thank you

